

Goal 1: Annotation Integration & Unified Chimpanzee Transcriptome

Objective Merge the major public chimpanzee annotations (RefSeq and Ensembl) into a single, non-redundant reference set, standardizing chromosome nomenclature and quantifying structural agreement to establish a baseline for population-scale isoform indexing.

Methods

- **Standardization:** Programmatically mapped NCBI accession identifiers to Ensembl standard nomenclature across GTFs and the mPanTro3 reference genome to prevent false-negative overlap.
- **Merging Strategy:** Processed annotations through IsoMatch using two stringency models:
Strict: Exact match required for splice junctions, Transcription Start Sites (TSS), and Transcription End Sites (TES).
Relaxed: Match required for internal splice junctions only (`-t none -T none`).

Key Finding 1: Gene Loci Consensus vs. Isoform Divergence The databases show strong consensus on the physical locations of transcribed regions but disagree heavily on alternative splicing.

- **Shared Gene Loci:** 24,686
- **Unique Gene Loci:** 9,781 (RefSeq) / 27,733 (Ensembl)

Key Finding 2: The Impact of UTR Boundaries Even when the databases agree on the internal splicing of a transcript, they rarely agree on the exact 5' and 3' UTR boundaries.

- Under *strict* boundaries, only ~6,300 transcripts matched perfectly.
- Under *relaxed* boundaries, **16,134 multi-exon transcripts** matched perfectly, proving that UTR definitions dictate a massive portion of the transcript-level divergence.

Key Finding 3: The Mono-Exon and Non-Coding Divide Ensembl annotates significantly more unique loci than RefSeq, driven almost entirely by single-exon, non-coding elements.

- Of Ensembl's unique genes, >85% are lncRNAs or misc_RNAs.
- Stratifying the relaxed transcript merge by exon count reveals **21,041 unique mono-exon transcripts in Ensembl**, compared to just 3,432 in RefSeq. RefSeq's pipeline requires substantially higher evidence thresholds to annotate single-exon non-coding elements, leaving Ensembl as the more aggressive, comprehensive database for pervasive transcription.

Deliverables:

1. The Core Reference Data (For Goals 2 & 3)

These are the files the rest of your team will actually plug into their pipelines.

- **merged_annotation.gtf.merged.gtf.gz**: This is your crown jewel. It's the unified transcript set with the **ISOM_SRC** tags.
- **merged_annotation.gtf.track.tsv.gz**: The tracking file that maps the new **ISOMT_** IDs back to their original RefSeq or Ensembl IDs.
- **genome_norm.fa** & **genome_norm.fa.fai**: The standardized reference genome. Sharing this ensures the whole team uses the exact same chromosome naming convention you established.

2. The Analytical Results (The Report)

These prove what you found and validate the quality of the merge.

- **transcript_overlap_summary.png**: Visual proof of the transcript-level divergence.
- **gene_overlap_summary.png**: Visual proof of the gene-level consensus.
- **A README.md file**: The written summary report we generated in the previous step explaining the methods and key findings.

3. The Codebase (For Reproducibility)

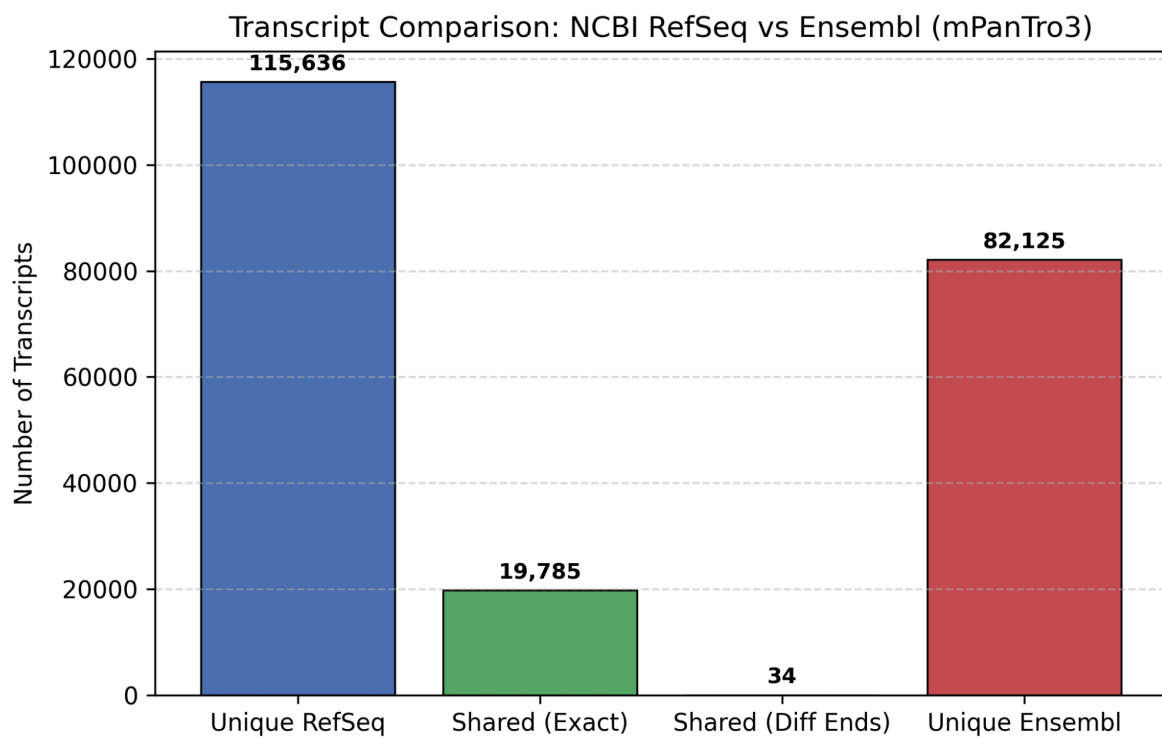
Provide your Python scripts so anyone can reproduce your stats.

- **analyze_plots.py**
- **analyze_genes.py**
- **analyze_exons.py**
- **analyze_biotypes.py**
- **analyze_refseq_biotypes.py**
- **GCF_028858775.2.chromAlias.txt** (The dictionary used to fix the chromosomes).

Results

Transcript based:

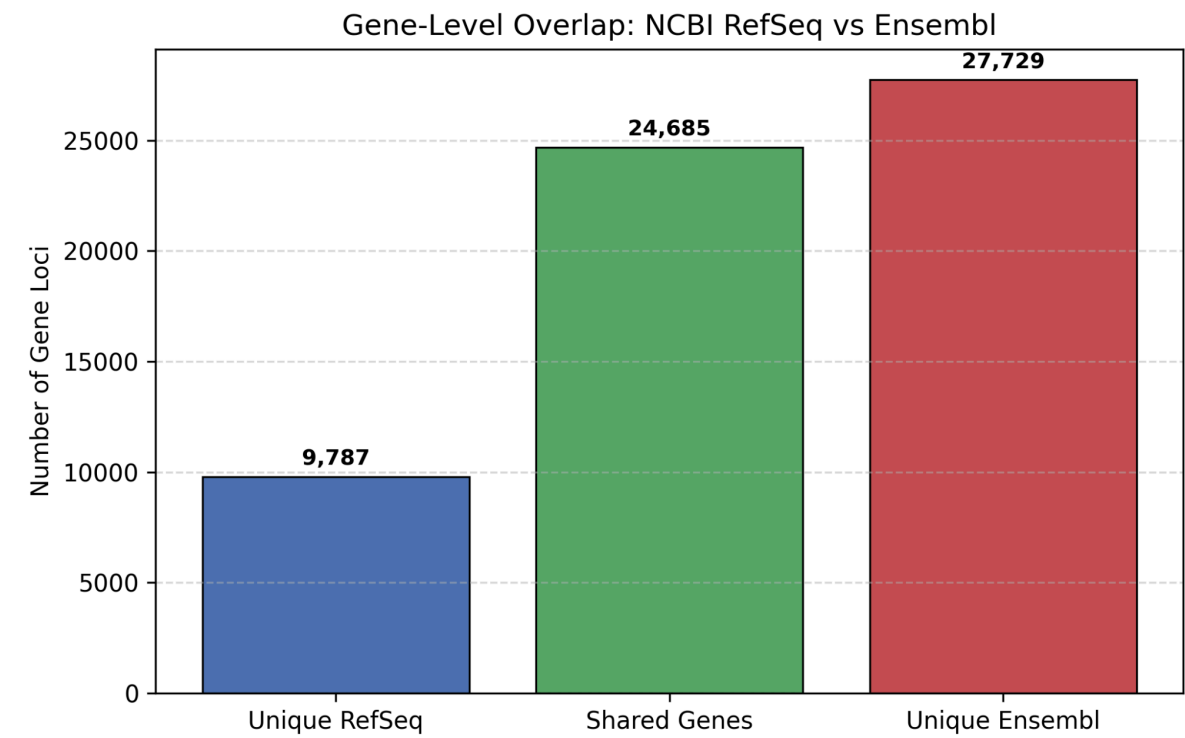
Transcript Annotation Metric	Count
Total RefSeq Transcripts	135455
- Multi-exon	128338
- Mono-exon	7117
Total Ensembl Transcripts	101944
- Multi-exon	77218
- Mono-exon	24726
Shared Transcripts	19819
- Exact Splice & Boundary Match	19785
- Structural Match / Boundary Diff	34
Unique to RefSeq	115636
Unique to Ensembl	82125



Gene Based:

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Gene Annotation Metric	Count
=====	
Total RefSeq Genes	34472
Total Ensembl Genes	52414

Shared Genes (Overlapping Loci)	24685
Unique to RefSeq	9787
Unique to Ensembl	27729
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Transcript Overlap (Splice Junction Only)	Count
=====	
** MULTI-EXON TRANSCRIPTS (High Confidence) **	
Shared	16134
Unique to RefSeq	112204
Unique to Ensembl	61084

** MONO-EXON TRANSCRIPTS (Lower Confidence) **	
Shared	3685
Unique to RefSeq	3432
Unique to Ensembl	21041
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Biotypes:

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Biotype of Unique Ensembl Genes	Count
=====	
lncRNA	18557
misc_RNA	5384
protein_coding	1694
Y_RNA	751
snRNA	443
miRNA	432
snoRNA	224
IG_V_gene	60
TR_V_gene	58
pseudogene	48
rRNA	33
TR_J_gene	24
vault_RNA	9
processed_pseudogene	6
ribozyme	4
TR_C_gene	2
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Biotype of Unique RefSeq Genes	Count
=====	
lncRNA	6155
protein_coding	2834
tRNA	505~6,300
miRNA	173
transcribed_pseudogene	43
V_segment	26
misc_RNA	24
rRNA	20
C_region	4
snRNA	3
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